

LUNA ELLIOTT

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Website: Luna's Website

Publications may appear under the name Luke Elliott.

RESEARCH PROFILE

Heilbronn Fellow at the University of Manchester working in topological algebra, geometric group theory and semigroup theory. Author of 19 research papers, including publications in Transactions of the American Mathematical Society, Advances in Mathematics, Journal of Algebra and Mathematics of Computation.

SELECTED PAPERS (10 OF 19)

See the complete list here.

- Automatic continuity, unique Polish topologies, and Zariski topologies on monoids and clones. L Elliott, J Jonušas, Z Mesyan, J Mitchell, M Morayne, Y Peresse. Transactions of the American Mathematical Society 376 (11), 8023-8093 2023
- Sufficient conditions for a group of homeomorphisms of the Cantor set to be two-generated. C Bleak, L Elliott, J Hyde. Journal of the Institute of Mathematics of Jussieu 23 (6), 2825-2858 2024
- A short proof of Rubin's theorem. J Belk, L Elliott, F Matucci. Israel Journal of Mathematics 267 (1), 157-169 2025
- Polish topologies on endomorphism monoids of relational structures. L Elliott, J Jonušas, JD Mitchell, Y Péresse, M Pinsker. Advances in Mathematics 431, 109214 2023
- E-disjunctive inverse semigroups. L Elliott, A Levine, J Mitchell. Journal of Algebra. 2025
- Topological embeddings into transformation monoids. S Bardyla, L Elliott, J Mitchell, Y Péresse. Forum Mathematicum 36 (6), 1537-1554 2024
- A description of $\text{Aut}(dVn)$ and $\text{Out}(dVn)$ using transducers. L Elliott. Groups, Geometry & Dynamics 17 (1), 2023
- The Diophantine problem in Thompson's group F. L Elliott, A Levine Mathematics of Computation. 2025
- Counting monogenic monoids and inverse monoids. L Elliott, A Levine, JD Mitchell Communications in Algebra 51 (11), 4654-4661 2023
- The Zariski Topology on Homeomorphism groups. L Elliott. arXiv:2601.15185 2026

DEGREES

2018-2021 St Andrews – PhD Pure Mathematics
Thesis: On constructing topology from algebra

ACADEMIC SERVICE

Co-organiser, Manchester Semigroup Seminar.
Referee for peer-reviewed mathematics journals.

This document was last updated on June 30th 2026

EMPLOYMENT HISTORY

- 2024-present: Heilbronn Fellow based at the University of Manchester.
- 2022-2024: Postdoctoral position at Binghamton University, New York State.
- 2022: Research Fellow, University of St Andrews, Programming (in GAP) algorithms for working with the GNS rational groups, automorphisms of the shift and Thompson groups.
- 2018-2021: PhD Student at the University of St Andrews
- 2017: Laidlaw Undergraduate Research and Leadership Program: Algorithms for commutative semigroups

TEACHING

See more teaching information here.
2022-2024: Lecturer. Writing and marking of exams homeworks and quizzes. Binghamton University.
2018-2021: Tutorials at University of St Andrews

SELECTED TALKS (10 OF 26)

See the complete list here.

- Thompson Monoids: RIMS conference (Kyoto) 2026
- Thompson Partition Monoids and Jónsson-Tarski algebras: TCA (Guimarães) 2026
- Topologizing Endomorphism Monoids: AAA (Vienna) 2026
- Semigroup Theory: Algebra seminar (Binghamton) 2024
- Automorphic Growth: Semigroup seminar (Manchester) 2026
- Thompson Monoids: NBSAN (Durham) 2025
- The rational group and automorphisms of Thompson's group V: Semigroup seminar (Manchester) 2025
- Epic Groups: Himr seminar (Manchester) 2025
- Finding topologies for semigroups: Semigroups, Automata and Languages seminar (Porto) 2024
- E-disjunctive inverse semigroups: NBSAN (Manchester) 2024

INTERESTS

Maths, Programming, Climbing and Japanese.

REFERENCES

Available upon request.